

LINGHAI LIU

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EDUCATION

Yale University	2023 - Present
Ph.D. in Statistics & Data Science	
Brown University	2019 - 2023
B.S. in Applied Mathematics - Computer Science (<i>Honors</i>) B.A. in Mathematics	
Magna Cum Laude Sigma Xi Senior Prize in Computer Science	

SELECTED RESEARCH EXPERIENCES

Composite Log-Concave Sampling 📄	Sept 2025 - Present
<i>Advisor: Sinho Chewi</i>	<i>Yale University</i>
- Improve the dimensional dependence of high-accuracy sampling for composite log-concave densities by implementing a restricted Gaussian oracle with independent Metropolis-Hastings proposals. - Analyzing the convergence of the independent Metropolis-adjusted Markov chain and Gibbs sampling scheme using conductance arguments by controlling higher-order Rényi divergence. - Liu, Linghai, and Sinho Chewi. A proximal gradient algorithm for composite log-concave sampling. <i>arXiv preprint arXiv:2605.12461 (2026)</i>.	
Topological Representations for Neural Networks	May 2022 - May 2023
<i>Advisor: Stuart Geman</i>	<i>Brown University</i>
- Designed generative models capable of learning a richer collection of abstract relationships using a multinomial idealization of biological neurons while ensuring model hierarchy and reusibility. - Developed sufficient statistics that exploit patterns of neuronal activities with topological structures such as cycles that can potentially be used to represent relationships.	
Jacobian-free Backpropagation in Inverse Problems 📄 🔄	May 2022 - Jan 2024
<i>Advisor: Samy Wu Fung</i>	<i>REU Researcher, Emory University</i>
- Compared classical optimization, standard feed-forward networks (Denoising CNN), Deep Unrolling methods, and implicit deep learning with applications to inverse problems in imaging. - Trained implicit networks to denoise images using Jacobian-free Backpropagation with fixed memory costs, which yielded comparable results with state-of-the-art models. - Liu, Linghai, Shuaicheng Tong, and Lisa Zhao. Training Implicit Networks for Image Deblurring using Jacobian-Free Backpropagation. <i>arXiv preprint arXiv:2402.02065 (2024)</i>.	

TEACHING & MENTORING

Teaching Fellow at Yale University	
· S&DS4320/S&DS6320: Advanced Optimization Techniques (TF)	Spring 2026
· S&DS4000/S&DS6000/MATH3300: Advanced Probability (TF)	Fall 2025
· S&DS242/S&DS542/MATH242: Theory of Statistics (TF)	Spring 2025
· S&DS230/S&DS530/ENV757/PLSC530: Data Exploration and Analysis (TF)	Fall 2024
Teaching Assistant at Brown University (*graduate level)	
· CSCI1952Q: Algorithmic Aspects of Machine Learning (TA)	Spring 2023
· APMA1660: Statistical Inference II (TA)	Spring 2023
· APMA1690: Computational Probability & Statistics (TA)	Fall 2022
· APMA1740/2610*: Recent Applications in Probability & Statistics (TA)	Spring 2022
· DATA1010*: Probability, Statistics and Machine Learning (TA)	Fall 2021
· CSCI1470/2470*: Deep Learning (Head TA)	Fall 2021

- APMA1650: Statistical Inference I (TA) Spring 2021
- **Peer Advisor, Meiklejohn Peer Advising Program** 2020 - 2021
- Advised 5 freshmen on academic and career development jointly with a faculty member.

HONORS & AWARDS

Emory REU/RET Computational Mathematics for Data Science, Emory University	2022
Second place, Henry Parker Manning Mathematical Prizes, Brown University	2022
BrownConnect SPRINT Undergraduate Teaching & Research Awards, Brown University	2021
Second place (tied), Hartshorn-Hypatia Freshman Math Contest, Brown University	2019

TECHNICAL STRENGTHS

Computer Languages	Python, R, Matlab, LaTeX, Java, C
Languages	English, Chinese, French